

Nachiket Karve

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Research Interests – Nonlinear Dynamics, Chaos Theory, Statistical Mechanics

EDUCATION

Boston University

PhD, Physics

2021 – Present

Advisors: [Dr David Campbell](#), [Dr Anatoli Polkovnikov](#)

Indian Institute of Technology, Kanpur

BS-MS Dual Degree, Physics

2016 – 2021

GPA: 9.6/10.0

Advisors: [Dr Loganayagam R](#), [Dr Nilay Kundu](#)

Thesis: Non-Polynomial Divergences in Wightman Correlation Functions

INVITED ARTICLES

- [1] [N. Karve](#), N. Rose, H. Kim, P. Mehta, D. K. Campbell, and A. Polkovnikov. *Understanding Chaos through Noise*. Commissioned Colloquium Article, Reviews of Modern Physics (invited; manuscript in preparation).

PAPERS AND PUBLICATIONS

- [1] [N. Karve](#), N. Rose, D. Campbell, and A. Polkovnikov. *Speed-Fisher Information: Chaos and Irreversibility in Classical and Quantum Dynamics*. 2026. arXiv: [2609.35687](#) [[cond-mat.stat-mech](#)]. URL: <https://arxiv.org/abs/2609.35687>.
- [2] [N. Karve](#), N. Rose, D. Campbell, and A. Polkovnikov. *Universal Dynamical Response to Slow Driving in Chaotic Systems*. 2026. arXiv: [2606.23810](#) [[cond-mat.stat-mech](#)]. URL: <https://arxiv.org/abs/2606.23810>.
- [3] N. Rose, [N. Karve](#), and D. K. Campbell. *Gradient of the Adiabatic Gauge Potential in Classical Systems*. 2025. arXiv: [2508.03804](#) [[nlin.CD](#)]. URL: <https://arxiv.org/abs/2508.03804>.
- [4] [N. Karve](#), N. Rose, and D. Campbell. *Diffusion as a Signature of Chaos*. 2025. arXiv: [2507.18617](#) [[nlin.CD](#)]. URL: <https://arxiv.org/abs/2507.18617>.
- [5] [N. Karve](#), N. Rose, and D. Campbell. “Adiabatic gauge potential as a tool for detecting chaos in classical systems”. In: *Phys. Rev. Res.* 7 (3 July 2025), p. 033006. DOI: [10.1103/1z5x-j644](#). URL: <https://link.aps.org/doi/10.1103/1z5x-j644>.

- [6] N. Karve, N. Rose, and D. Campbell. “Periodic orbits in Fermi–Pasta–Ulam–Tsingou systems”. In: *Chaos: An Interdisciplinary Journal of Nonlinear Science* 34.9 (Sept. 2024), p. 093117. ISSN: 1054-1500. DOI: [10.1063/5.0223767](https://doi.org/10.1063/5.0223767). eprint: <https://pubs.aip.org/aip/cha/article-pdf/doi/10.1063/5.0223767/20161085/093117\1\5.0223767.pdf>. URL: <https://doi.org/10.1063/5.0223767>.
- [7] N. Karve and R. Loganayagam. *Heisenberg Picture for Open Quantum Systems*. 2020. arXiv: [2011.15118](https://arxiv.org/abs/2011.15118) [quant-ph]. URL: <https://arxiv.org/abs/2011.15118>.

TALKS AND PRESENTATIONS

Universal Dynamical Response to Slow Driving in Chaotic Systems, *Penn State University, Jun 2026* [poster]

Understanding Chaos through Physical Observables, *University of Arizona, Jan 2026* [poster]

Chaos in Classical Systems, *Boston University, May 2025* [slides]

Chaos in FPUT-like Systems, *March Meeting 2025, Anaheim* [slides]

The Metastable State of the FPUT Problem, *March Meeting 2024, Minneapolis* [slides]

The Metastable State of the FPUT Problem, *March Meeting 2023, Las Vegas* [slides]

AWARDS AND HONORS

Dynamics Days Travel Grant *Jan, 2026*

KVPY (Kishore Vaigyanik Protsahan Yojana) Fellowship *2016 – 2021*

Academic Excellence Award, IIT Kanpur *2016, 2017, 2018, 2019*

Silver Medal, University Physics Competition *2017*

TEACHING EXPERIENCE

Boston University *2021 – Present*

Quantum Computing (PY 536), Teaching Fellow and Guest Lecturer

General Physics I (PY 211), Grader

General Physics II (PY 212), Teaching Fellow and Grader

Physics I (PY 105), Teaching Fellow

Physics II (PY 106), Teaching Fellow

MENTORSHIP

Undergraduate Students

Emir Akdag

Jonah Gluck

Kristen Bestavros

ACADEMIC SERVICES

Physical Review Referee	<i>2025 – Present</i>
Head and Volunteer at Prayas , IIT Kanpur's Community Welfare School	<i>2018 – 2021</i>
Academic Mentor, IIT Kanpur	<i>2017 – 2018</i>